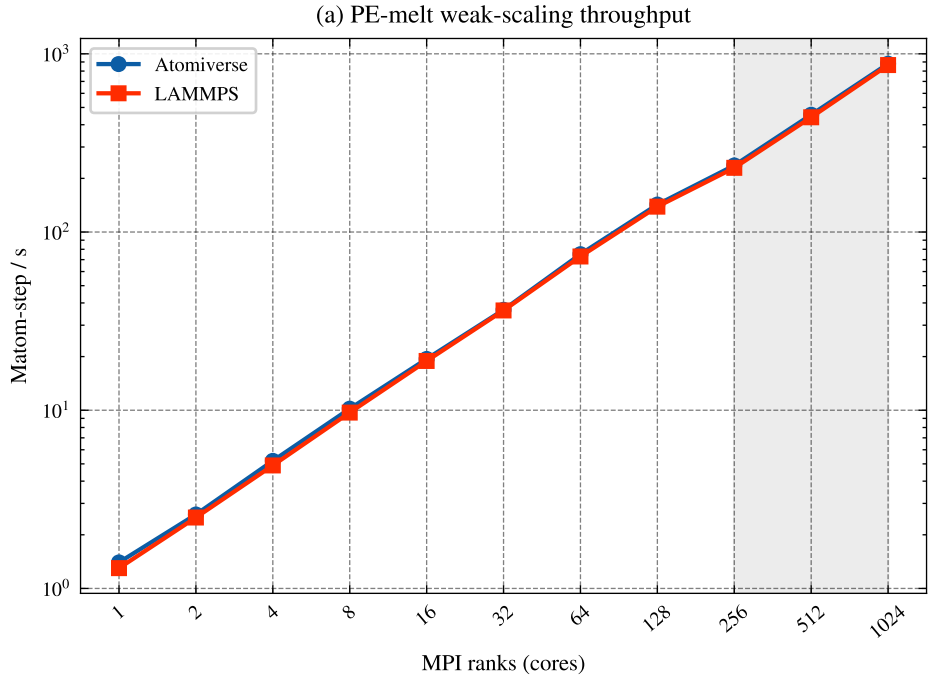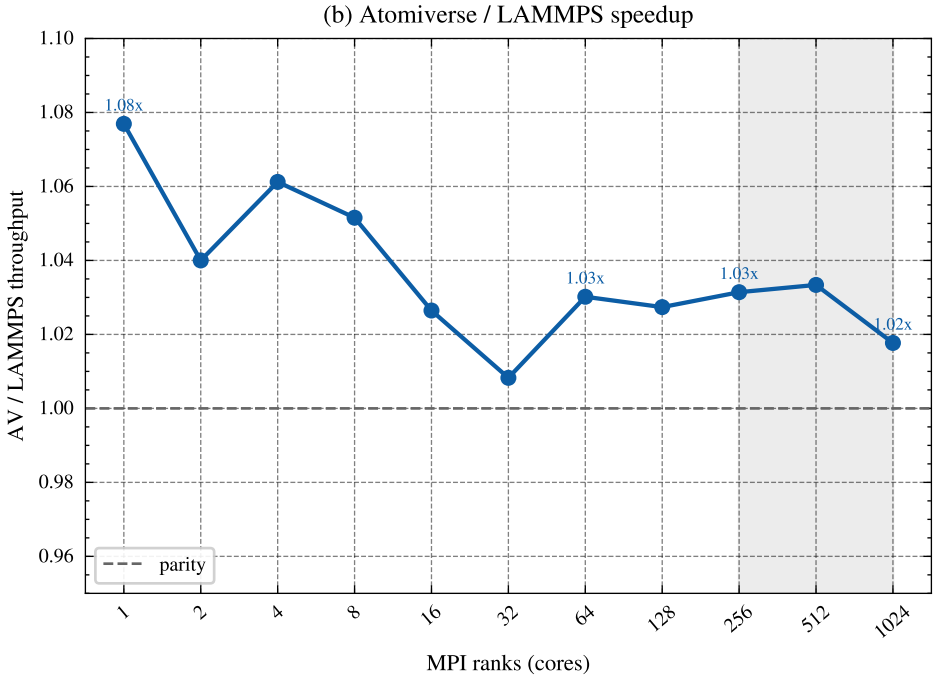Atomiverse vs LAMMPS — PE-melt CPU weak-scaling after force direct-write + reduce-cadence opt (shaded = cross-node). AV now leads 1.02-1.06x at every scale (was 0.92x @256).